Quadratic maps satisfying $f \circ f = f + \mathrm{id}$: polar Lie structure and sharp dimension bounds

Abstract

The equation $f \circ f = f + \mathrm{id}$ couples three-cycles to vector addition: each nonzero orbit consists of the three nonzero points of a binary plane. We study its solutions of binary algebraic degree at most two through their polar Lie algebras. The equation forces both the Jacobi identity and an additional degree-four identity, and every solvable polar algebra is nilpotent. In the metabelian case we obtain a complete normal form and an equivalence, including isomorphisms and automorphism groups, with metabelian $\mathbb{F}_4$-Lie algebras whose quotient actions on the derived ideal have square zero. The correspondence gives the exact minimum binary dimension at every polar nilpotency class $c \geq 2$: $6, 8, 12$ in classes two, three, and four, and $2^c + 2c - 6$ for $c \geq 5$. We characterize when the original polar bracket admits an $\mathbb{F}_4$-bilinear structure extending its intrinsic scalar actions. In this compatible subclass the minimum is $2^c + 2c - 2$ for every $c \geq 2$, with at least $c - 2$ inequivalent extremizers for every $c \geq 4$. Two closed classifications illustrate the structure theory: all eight-dimensional solutions form three linear-conjugacy classes, and the solvable solutions with four-dimensional binary abelianization form an exhaustive chain of seven classes ending in dimension sixteen. Exact central-extension certificates make the finite rank calculations in the latter classification independently reproducible.

Keywords. Quadratic map; complete permutation; polar Lie algebra; metabelian Lie algebra; restricted Lie algebra; nilpotency class; finite field.

1 Introduction

Let V be a finite-dimensional vector space over $\mathbb{F}_2$. We consider the functional equation

$$f(f(x)) = f(x) + x \qquad (x \in V) \tag{1}$$

under the assumption that f has binary algebraic degree at most two. Every nonzero orbit is the triple $\{x, f(x), x + f(x)\}$, the nonzero points of a binary plane. The equation therefore ties a local cycle condition to the additive geometry of the whole space. We ask how a quadratic degree bound organizes these orbits into an algebraic structure that can be classified and measured.

The linear case identifies the relevant scalar field. A linear solution has minimal polynomial $X^2 + X + 1$ on every nonzero space, so it defines an intrinsic $\mathbb{F}_4$-structure in which f is multiplication by ω, where $\omega^2 + \omega = 1$. Up to binary-linear conjugacy, there is one such solution in each positive even dimension. Degree two is the first level at which the polar form can be nonzero. It introduces interactions between the orbit planes, and these interactions are the subject of the paper.

The first structural object is the polar form

$$[x, y] = f(x + y) + f(x) + f(y).$$

For a general zero-constant quadratic map this form is alternating and bilinear. Here the functional equation also forces Jacobi, making it a Lie bracket. The derived space, the center,

and the lower central series therefore become intrinsic invariants of the map. The equation imposes further rigidity: solvability implies nilpotency, and fourth polarization gives a degree-four polynomial identity. This polar structure is the bridge from the permutation equation to both classification and sharp dimension bounds.

1.1 The main correspondence

Write $W = [V, V]$ for the binary span of all polar values. The principal classification theorem concerns the metabelian case $[W, W] = 0$. The maps induced by f on W and V/W are linear operators satisfying $T^2 + T = \mathrm{id}$; they therefore define intrinsic $\mathbb{F}_4$-vector space structures. Fix $\omega \in \mathbb{F}_4$ with $\omega^2 + \omega = 1$.

In suitable binary-linear coordinates, every metabelian solution has the form

$$f(x, z) = \big(\omega x,\, \omega z + P(x)z + Q(x) + P(x)Q(x)\big), \tag{2}$$

where

$$P(x) = \sum_{i=1}^{d} x_i P_i, \qquad Q(x) = \sum_{i<j} x_i x_j a_{ij}.$$

The $\mathbb{F}_4$-linear operators P_i commute, satisfy $P_i^2 = 0$, and obey

$$P_i a_{jk} + P_j a_{ik} + P_k a_{ij} = 0.$$

The coefficients a_{ij} generate W under the operators P_i. Conversely, all such data give solutions.

The key necessity statement is that binary degree two forces the square-zero actions on the intrinsic polar image W. Thus the normal form describes every metabelian solution as well as constructing them. Moreover, the correspondence includes morphisms. Binary-linear conjugacy of these maps is equivalent to $\mathbb{F}_4$-Lie isomorphism of the associated metabelian algebras with $(\operatorname{ad} u)^2|_W = 0$. In particular it identifies the linear centralizer of f with the automorphism group of the associated $\mathbb{F}_4$-Lie algebra (Theorem 3.2).

1.2 Sharp dimension costs

The normal form gives an exact measure of the dimension required for a given polar Lie nilpotency class. Let $\mu(c)$ be the minimum binary dimension among metabelian solutions of class $c \geq 2$. We prove

$$\mu(2) = 6, \qquad \mu(3) = 8, \qquad \mu(4) = 12, \qquad \mu(c) = 2^c + 2c - 6 \quad (c \geq 5). \tag{3}$$

The proof uses a universal commutator module over a square-zero polynomial ring. Its multigrading determines the possible terminal commutators, and a dual contraction argument determines the smallest quotient that retains one of them. Both the lower bound and its attainment are uniform in c (Theorem 4.4).

The $\mathbb{F}_4$-Lie algebra in this correspondence is auxiliary: the original binary polar bracket need not itself be $\mathbb{F}_4$-bilinear. We give an exact linear criterion for when it admits such a structure with the prescribed intrinsic scalar actions on W and V/W. This condition is equivalent to the existence of a restricted square on the auxiliary algebra that vanishes on W and takes values in W (Theorem 5.2).

Let $\nu(c)$ denote the minimum dimension in this compatible subclass. We obtain the single formula

$$\nu(c) = 2^c + 2c - 2 \qquad (c \geq 2). \tag{4}$$

One part of its lower bound counts independent preimages of the nonempty faces of a square-zero operator cube. This gives $\dim_{\mathbb{F}_4} W \geq 2^{c-1}-1$, independently of the unrestricted extremal argument. The other part proves $\dim_{\mathbb{F}_4}(V/W) \geq c$. The elementary augmentation-module construction attains both bounds (Theorem 6.3).

Table 1: Minimum binary dimensions in the two metabelian subclasses.

Polar class c	2	3	4	5	6	7
$\mu(c)$	6	8	12	36	70	136
$\nu(c)$	6	12	22	40	74	140

Compatibility therefore has an exact additional cost of four binary dimensions in every class $c \geq 5$. Equality still allows growing structural diversity: for each $c \geq 4$ we construct at least $c-2$ pairwise inequivalent compatible extremizers (Theorem 6.4).

1.3 Closed classification ranges

Two applications illustrate the reach and the boundaries of the theory. There are exactly three binary-linear conjugacy classes in dimension eight (Theorem 7.4). The proof combines the correspondence with the classical small-dimensional simple Lie algebra lists and short explicit obstructions to the polar identities.

There is also a closed classification in an all-dimensional stratum: if the polar algebra is solvable and $\dim_{\mathbb{F}_2}(V/W) = 4$, exactly one class occurs in each of dimensions $4, 6, 8, 10, 12, 14, 16$, and none occurs thereafter (Theorem 7.5). Its final two members are nonmetabelian. Completeness is proved by a lossless central descent, exact extension-space calculations, and explicit projective orbit actions. The finite rank portion is supplied with independently replayable certificates.

1.4 Context and related work

There is an operator viewpoint on (1). On complete mappings, $\mathcal{F}(g) = (\mathrm{id}+g)^{-1}$ satisfies $\mathcal{F}^3(g) = g$, and its fixed objects are exactly the solutions of (1). The arithmetic study of norm fibers and affine companion groups in [4] also describes these fixed objects by oriented binary line spreads. The present paper studies their quadratic structure through the polar Lie algebra.

Equation (1) implies that both f and $f+\mathrm{id}$ are permutations. Triple-cycle and regular complete permutation polynomials have been studied by Liu–Chen–Xu–Sun [5], Xu–Zeng–Zhang [13], and Lu–Wu–Wang–Cao [7]. Quadratic complete mappings also occur in the constructions of Samardjiska–Gligoroski [10]. Here the orbit-sum identity, binary degree, and binary-linear conjugacy are considered together. In particular, simultaneous three-cycle behavior of f and $f+\mathrm{id}$ does not imply (1); Example 2.7 gives a quadratic instance from the affine-switching construction of [13].

There is a parallel quasigroup interpretation. The operation $x * y = x + f(x+y)$ is a quaternary Mendelsohn quasigroup. Zhang's thesis [14, Chapter 3] develops central representations and recursions in that setting. Its Lemma 10.3(1) already contains the central six-dimensional example used below. Throughout this paper, equivalence means binary-linear conjugacy, and nilpotency class means the class of the polar Lie algebra. These specifications retain the addition and the degree filtration that enter the necessity proofs.

The commutator-module presentation is the classical Koszul presentation of the free metabelian derived module; see Papadima–Suciu [9, Theorem 6.2] and the characteristic-two module methods of Kovács–Stöhr [2]. Restricted squares and quadratic maps have a broader

established connection; see, for example, Pakianathan–Yalçın [8]. We use these tools to prove the degree-forced correspondence and the exact extremal formulas for (1). The four-variable polar identity is the classical degree-four standard polynomial in Lie form [1, Theorem 9]; its forced occurrence here provides the obstructions used in the low-dimensional application.

1.5 Conventions

We use f^j only for composition powers. All dimensions without a specified field are binary dimensions. A map is called quadratic when its binary algebraic degree is at most two, so the linear cases are included. Two solutions are equivalent if $Sf = f'S$ for a binary-linear isomorphism S. Lie brackets are alternating; in characteristic two they are symmetric as bilinear maps. We put $\gamma_1(L) = L$, $\gamma_{i+1}(L) = [L, \gamma_i(L)]$, and $L^{(0)} = L$, $L^{(i+1)} = [L^{(i)}, L^{(i)}]$. The class of a nonzero abelian algebra is one.

Notation at a glance. The $\mathbb{F}_4$-dimensions below are used in the metabelian correspondence.

Symbol	Meaning
$L = (V, [\ ,\ ])$	The binary polar Lie algebra of f.
$W = L',\ U = V/W$	The intrinsic derived ideal and abelianization.
$d,\ w$	$\dim_{\mathbb{F}_4} U$ and $\dim_{\mathbb{F}_4} W$.
ω	A root of $X^2 + X + 1$ in $\mathbb{F}_4$.
$\mathfrak{m}$	The augmentation ideal $(t_1, \dots, t_d)$ of R_d, or its image under the operator action.
$\gamma_i(L),\ L^{(i)}$	The lower central and derived series.
$\mu(c),\ \nu(c)$	Minimum binary dimensions at class c: metabelian and compatible metabelian, respectively.
$\mathcal{A}_k$	Filtered action spaces used to distinguish extremizers in Theorem 6.4.

2 The polar Lie algebra

Proposition 2.1. *Every solution of* (1) *satisfies*

$$f^3 = \mathrm{id}, \qquad f(0) = 0, \qquad f(x + f(x)) = x.$$

Its only fixed point is zero. Every nonzero orbit spans a binary plane, and these planes partition $V \setminus \{0\}$. *In particular* $\dim V$ *is even.*

Proof. Evaluate (1) at $f(x)$. It gives $f^3(x) = f^2(x) + f(x) = x$, so f is bijective. If $a = f(0)$, then $f(a) = a$; applying f^3 at zero gives $a = 0$. A fixed point in (1) must be zero. The remaining identity follows from $x + f(x) = f^2(x)$. The nonzero orbits have length three and sum zero, and $3 \mid 2^{\dim V} - 1$ implies the parity assertion. □

Theorem 2.2. *Let* f *be a quadratic solution. Its polar form* $[x, y] = f(x + y) + f(x) + f(y)$ *is a Lie bracket. It satisfies*

$$[x, f(x)] = 0, \tag{5}$$

$$f([x, y]) = [f(x), f(y)] + [x, y] + [f(x) + f(y), [x, y]], \tag{6}$$

$$[[a, b], [c, d]] + [[a, c], [b, d]] + [[a, d], [b, c]] = 0. \tag{7}$$

Proof. The polar is alternating and bilinear because f is quadratic and $f(0) = 0$. Expand $f(x + f(x)) = x$ by polarization. The terms $f(x) + f^2(x)$ sum to x, leaving (5). Take the third additive difference at zero of $x \mapsto [x, f(x)]$. The terms involving individual values of f cancel, leaving

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

This is Jacobi.

Next polarize $f^2 = f + \mathrm{id}$. Since $f(x+y) = f(x) + f(y) + [x, y]$, the expansion of the left side gives (6).

For (7), introduce Boolean variables $t_1, \ldots, t_4$ and vectors $x_1, \ldots, x_4$. The exact expansion is

$$f\Big(\sum_i t_i x_i\Big) = \sum_i t_i f(x_i) + \sum_{i<j} t_i t_j [x_i, x_j].$$

In applying f again, the coefficient of $t_1 t_2 t_3 t_4$ can only come from a polar cross term between two disjoint pair summands. There are three such pairings, giving the left side of (7). The composition has degree at most two, so this coefficient vanishes. All coefficient comparisons take place in the Boolean function algebra, with $t_i^2 = t_i$. □

The two polarization steps have distinct roles. Binary degree two gives an alternating bilinear form; the commuting identity (5) forces its Jacobi identity. The fourth polarization then supplies (7), an additional restriction on the resulting Lie algebra. The polynomial in (7) is the associative standard polynomial of degree four expressed as a Lie element in characteristic two [1, Theorem 9]. Its forced occurrence here gives the explicit exclusions in Section 7.

Proposition 2.3. *All terms of the derived, lower central, and upper central series of the polar Lie algebra $L = (V, [\ ,\ ])$ are f-invariant. The center is f-invariant and f is linear on it. Every nonzero factor of any of these series has even binary dimension.*

Proof. Let I be an f-invariant ideal. In (6), take $x \in I$. Each term on the right belongs to $[I, L]$, which is an ideal. Quadratic expansion extends invariance from bracket generators to their span: the extra polar terms also lie in that ideal. The same argument with $x, y \in I$ proves invariance of $[I, I]$, using that it is an ideal of L. Induction starts with $I = L$.

Polarization of (5) gives

$$[x, f(y)] + [y, f(x)] + [x, [x, y]] + [y, [x, y]] = 0. \tag{8}$$

If $x \in Z(L)$, this says $[y, f(x)] = 0$ for every y. Thus $Z(L)$ is invariant; its polar is zero, so f is linear there. Pass successively to quotients by invariant central ideals to obtain the upper central assertion. Every series factor inherits a solution of (1), so Proposition 2.1 proves its parity. □

Lemma 2.4 (Ideal descent)**.** *For every Lie ideal I of L,*

$$\{f(x) + I : x \in I\} \subseteq Z(L/I).$$

Every perfect ideal is f-invariant. Every ideal I for which $Z(L/I) = 0$ is also f-invariant.

Proof. For $x \in I$, all terms in (8) except $[y, f(x)]$ belong to I. This proves the inclusion. The map $\phi : I \to L/I$, $x \mapsto f(x) + I$, is binary-linear because its polar takes values in I. Equation (6) gives $\phi([x, y]) = [\phi(x), \phi(y)]$ for $x, y \in I$. Its image is central, so $\phi([I, I]) = 0$. This proves the perfect-ideal assertion. The centerless-quotient assertion follows immediately from the inclusion. Bijectivity and finiteness turn each invariant inclusion into equality. □

Corollary 2.5. *If a quadratic solution has a nonsolvable polar algebra, then there is a quadratic solution whose polar algebra is nonabelian simple, of even binary dimension no larger than the original dimension.*

Proof. The stabilized derived ideal P is nonzero and perfect, hence invariant. Choose a maximal proper Lie ideal M of P. The quotient P/M is nonabelian simple and centerless. Apply Lemma 2.4 inside P to see that M is invariant. Restriction and quotient therefore give the required quadratic solution. □

Lemma 2.6. *A nonabelian nilpotent polar Lie algebra has binary abelianization dimension at least four.*

Proof. The abelianization is a nonzero even-dimensional space with the linear action induced by f. If its dimension were two, any nonzero coset $x+L'$ and its image $f(x)+L'$ would form a binary basis. The subalgebra H generated by $x, f(x)$ is abelian by (5). Moreover $L = H + L'$. Inductively, $L = H + \gamma_j(L)$ implies $L = H + \gamma_{j+1}(L)$, because $[H + \gamma_j, H + \gamma_j] \subseteq H + \gamma_{j+1}$. Nilpotency then gives $L = H$, a contradiction. □

Example 2.7. Let $\mathbb{F}_{16}$ contain $\mathbb{F}_4$, let $\eta \in \mathbb{F}_4 \setminus \mathbb{F}_2$, and write $\chi(x) = 1$ off $\mathbb{F}_4$ and $\chi(x) = 0$ on $\mathbb{F}_4$. The affine-switching construction of [13, Corollary 6] includes

$$G(x) = \eta x + \chi(x) = \eta x + \mathrm{Tr}_{\mathbb{F}_{16}/\mathbb{F}_4}(x)^3.$$

The trace is binary-linear, and its cube is binary-quadratic. Both $\mathbb{F}_4$ and its complement are invariant under multiplication by η and under translation by one. Hence G is a permutation with $G^3 = \mathrm{id}$ and only zero fixed. The same holds for $G + \mathrm{id}$. Nevertheless

$$G^2(x) + G(x) + x = \eta\chi(x),$$

so G does not satisfy (1). This separates the equation studied here from simultaneous triple-cycle complete permutation behavior even within binary degree two.

3 A complete metabelian correspondence

Assume in this section that the polar Lie algebra is metabelian. Then $W = L'$ is abelian, $f|_W$ is linear, and the induced map on $U = V/W$ is linear. Both operators satisfy $T^2 + T = \mathrm{id}$. We give U and W the $\mathbb{F}_4$-structures in which these operators are multiplication by ω.

Lemma 3.1 (The operator pencil)**.** *In any binary-linear splitting of $V \to U$, write*

$$f(x, z) = (\omega x, \omega z + L_x z + q(x)). \tag{9}$$

The binary-linear pencil $x \mapsto L_x$ satisfies

$$L_x\omega = \omega L_x, \quad L_x^3 = 0, \quad L_{\omega x} = \omega L_x + L_x^2, \quad L_x L_y = L_y L_x, \quad L_x^2 L_y = 0.$$

Proof. Put $J = \omega\,\mathrm{id}_W$ and $M_x = J + L_x$. Both M_x and $\mathrm{id} + M_x$ are invertible: they are the restrictions on coset differences of the permutations f and $f + \mathrm{id} = f^2$. Substitution in (1) gives

$$M_{\omega x} M_x = \mathrm{id} + M_x.$$

For $A = M_x$, the three matrices along the ω-orbit are $A, \mathrm{id} + A^{-1}, (\mathrm{id} + A)^{-1}$. Their sum is J, because the three L-terms sum to zero. Thus J is a rational function of A, and in particular commutes with it. Clearing the denominators gives

$$JA(\mathrm{id} + A) = A^3 + A + \mathrm{id}\,.$$

Since $J^2 + J = \mathrm{id}$ and $J^3 = \mathrm{id}$, this is $(A + J)^3 = 0$. It follows that L_x commutes with J and is cube zero. Expanding $(J + L_x)^{-1}$ now gives $L_{\omega x} = JL_x + L_x^2$. The additivity of the left side forces the L_x's to commute.

For $P = L_x, Q = L_y$, the identity $(P + Q)^3 = 0$ yields $P^2Q = PQ^2 =: T$. Apply the same identity to $P' = JP + P^2 = L_{\omega x}$ and Q. It gives $(J^2 + J)T = P^2Q^2$, hence $T = P^2Q^2 = PT$. As P is nilpotent, this implies $T = 0$. □

Theorem 3.2 (Normal form and equivalence)**.** *Every metabelian quadratic solution is binary-linearly conjugate to*

$$f(x,z) = \big(\omega x, \omega z + P(x)z + Q(x) + P(x)Q(x)\big) \quad \text{on } \mathbb{F}_4^d \oplus W, \tag{10}$$

where

$$P(x) = \sum_i x_i P_i, \qquad Q(x) = \sum_{i<j} x_i x_j a_{ij}, \tag{11}$$

$$P_i P_j = P_j P_i, \qquad P_i^2 = 0, \tag{12}$$

$$P_i a_{jk} + P_j a_{ik} + P_k a_{ij} = 0 \quad (i<j<k), \tag{13}$$

$$W = \sum_i P_i W + \operatorname{span}_{\mathbb{F}_4}\{a_{ij} : i<j\}. \tag{14}$$

Conversely these conditions define a quadratic solution with intrinsic polar image W.

The binary-linear conjugacy groupoid of these maps is equivalent to the isomorphism groupoid of finite-dimensional metabelian $\mathbb{F}_4$-Lie algebras $\mathfrak{l}$ satisfying

$$(\operatorname{ad} u)^2|_{\mathfrak{l}'} = 0 \qquad (u \in \mathfrak{l}). \tag{15}$$

For corresponding objects, $C_{\mathrm{GL}(V,2)}(f) \cong \mathrm{Aut}_{\mathbb{F}_4}(\mathfrak{l})$. Their lower central dimensions agree after multiplication by two.

Proof. Lemma 3.1 provides the preliminary actions. We first normalize a section and separate the scalar weights of the cocycle (Steps 1–2). The decisive step is to combine binary degree two with intrinsic generation to force square zero (Step 3). We then verify sufficiency, construct the auxiliary Lie algebra and compare lower central series (Steps 4–5). Finally, tracking changes of coordinates upgrades the normal form to an equivalence on morphisms (Step 6).

Step 1: a normalized section. Choose an $\mathbb{F}_4$-basis e_i of U, with arbitrary lifts v_i in V. Define a binary-linear section by $e_i \mapsto v_i$, $\omega e_i \mapsto f(v_i)$. The images form a basis modulo W. Since $f(v_i + f(v_i)) = v_i$, the cocycle q in (9) vanishes on each coordinate $\mathbb{F}_4$-line.

Step 2: scalar weights. Initially put $P_i = L_{e_i}$, without yet asserting $P_i^2 = 0$. Lemma 3.1 gives

$$L_{\alpha e_i} = \alpha P_i + (\alpha + \alpha^2) P_i^2 \quad (\alpha \in \mathbb{F}_4).$$

Set $N(x) = \sum_i x_i(P_i + P_i^2)$. Then $L_x = N + N^2$, $N^3 = 0$, and $L_{\omega x} = \omega N + \omega^2 N^2$. Decompose $q = q_0 + q_1 + q_2$ into the three scalar weights $q_j(\omega x) = \omega^j q_j(x)$. Projection to each weight is a linear combination of $q(x), q(\omega x), q(\omega^2 x)$, so preserves binary degree and the vanishing on coordinate lines.

The cocycle equation is

$$q(\omega x) = \omega^2 q(x) + (\omega N + \omega^2 N^2) q(x).$$

Its weight components give

$$q_0 = N q_2 + \omega N^2 q_1, \qquad q_1 = \omega N q_0 + \omega^2 N^2 q_2.$$

Using $N^3 = 0$ yields

$$q_0 = N q_2, \qquad q_1 = N^2 q_2. \tag{16}$$

Step 3: binary degree forces square zero. The reduced $\mathbb{F}_4$-polynomial of a weight-two binary-quadratic map has the form $\sum_i b_i x_i^2 + \sum_{i<j} a_{ij} x_i x_j$. Line normalization removes the first sum, so $q_2 = Q$. Consequently

$$q_1 = N^2 Q = \sum_{k,i<j} x_k^2 x_i x_j P_k^2 a_{ij}.$$

This is where the degree bound becomes decisive. For $k = i$ or j, the exponent pattern is $(3, 1)$; otherwise it is $(2, 1, 1)$. Both patterns have binary degree three, whereas the weight projection q_1 has degree at most two. The cubed or squared coordinate identifies k, and the remaining coordinates identify $\{i, j\}$. Thus each displayed term has its own reduced monomial, forcing

$$P_k^2 a_{ij} = 0 \qquad \text{for all } k, i, j. \tag{17}$$

For completeness, the one-coordinate function basis $1, x, x^2, x^3$ has binary degrees $0, 1, 1, 2$. Its leading classes form a basis in the associated graded degree filtration. Tensoring over the coordinates proves that distinct reduced monomials of the displayed degree three cannot cancel to degree two. This justifies the comparison in the binary degree filtration.

We now extend this vanishing from the cocycle coefficients to the whole intrinsic derived ideal. Let $\mathfrak{m}W = \sum_i P_i W$. By the preliminary identities, every P_k^2 annihilates $\mathfrak{m}W$. Modulo $\mathfrak{m}W$, the polar values in (9) are exactly the values of the polar of Q. Since W is the span of all polar values, this proves (14). Together with (17), it gives $P_k^2 = 0$ on all of W. Now $L_x = P(x)$ and $q = Q + PQ$. The degree-three terms of PQ with three distinct indices vanish precisely when (13) holds. The other terms have exponent patterns $(2, 1)$, of binary degree two. This proves necessity.

Step 4: sufficiency. Conditions (12) and (13) make $P(x)^2 = 0$ and make PQ binary-quadratic. Under $x \mapsto \omega x$, the functions Q and PQ have weights two and zero. Thus

$$(Q + PQ)(\omega x) = \omega^2 Q + PQ = \omega^2 (Q + PQ) + \omega P(Q + PQ).$$

Together with $(\omega \,\mathrm{id} + \omega P)(\omega \,\mathrm{id} + P) = \omega^2 \,\mathrm{id} + P$, this verifies $f^2 = f + \mathrm{id}$ by substitution. Mixed polar values generate $\mathfrak{m}W$; modulo that space, the values of the polar of Q generate the a_{ij}'s over $\mathbb{F}_4$. Equation (14) therefore makes the polar image exactly W.

Step 5: the associated Lie algebra. Put $\beta(x, y) = Q(x + y) + Q(x) + Q(y)$ and define

$$[(x, z), (y, t)]_0 = (0, P(x)t + P(y)z + \beta(x, y)). \tag{18}$$

Jacobi follows from commuting operators and (13). This is a metabelian $\mathbb{F}_4$-Lie algebra with derived ideal W, and it satisfies (15). Conversely, splitting the abelianization of any algebra with these properties gives exactly such operators and an alternating cocycle. Choose its cross-term refinement Q to recover (10).

For both the original polar bracket and (18), the action on W is $P(x)$. Hence, with $\mathfrak{m}$ the ideal generated by these actions,

$$\gamma_2 = W, \qquad \gamma_k = \mathfrak{m}^{k-2} W \quad (k \geq 3). \tag{19}$$

This proves the lower central comparison.

Step 6: morphisms. A binary-linear conjugacy preserves the intrinsic space W. Its diagonal blocks $A : U \to U'$ and $C : W \to W'$ commute with the intrinsic scalar actions, hence are $\mathbb{F}_4$-linear. Write its off-diagonal block as $t_s + t_a$, where $t_s(\omega x) = \omega t_s(x)$ and $t_a(\omega x) = \omega^2 t_a(x)$. Direct substitution in $Sf = f'S$ gives the two conditions

$$CP(x) = P'(Ax)C, \tag{20}$$

$$Q'(Ax) = CQ(x) + t_a(x) + P'(Ax)t_s(x). \tag{21}$$

Indeed the unseparated cocycle equation reduces to the second one by multiplication with $\mathrm{id} + P'(Ax)$, whose inverse is itself. Polarization over $\mathbb{F}_4$ eliminates t_a and gives

$$\beta'(Ax, Ay) = C\beta(x, y) + P'(Ax)t_s(y) + P'(Ay)t_s(x). \tag{22}$$

Equations (20) and (22) say exactly that $(x, z) \mapsto (Ax, Cz + t_s(x))$ is an $\mathbb{F}_4$-Lie isomorphism.

Conversely, for any such Lie isomorphism the function $Q'(Ax) + CQ(x) + P'(Ax)t_s(x)$ has zero polar and is an ordinary homogeneous quadratic. It is therefore a unique sum of squares $\sum b_i x_i^2$. Taking this as t_a gives a unique conjugacy. Scalar parts respect composition because A and C are $\mathbb{F}_4$-linear. Thus the correspondence is an equivalence on both objects and morphisms, proving also the centralizer assertion. □

Corollary 3.3. *Every metabelian polar algebra is nilpotent and satisfies* $(\operatorname{ad} v)^3 = 0$ *for every* v. *If* $w = \dim_{\mathbb{F}_4} W$, *its class is at most* $w + 1$.

Proof. The commuting square-zero operators generate a nilpotent associative ideal. They have a common nonzero kernel whenever $W \neq 0$; repeating on the quotient gives a strictly triangular flag, so all products of w operators vanish. Apply (19). Finally $\operatorname{ad} v$ maps V into W, and its square on W is zero. □

Theorem 3.4. *A polar Lie algebra is solvable if and only if it is nilpotent. Its solvable radical is invariant under* f *and is nilpotent.*

Proof. Induct on derived length. The abelian case is immediate. Put $W = L'$, $W' = L''$. The restricted solution on W has nilpotent polar algebra by induction. The quotient L/W' is metabelian, hence nilpotent by Corollary 3.3. For fixed $x \in L$, the derivation $D = \operatorname{ad} x|_W$ is therefore nilpotent on W/W'.

We use the following elementary derivation fact. If N is nilpotent and D is nilpotent on N/N', then it is nilpotent on N. Each factor $\gamma_i(N)/\gamma_{i+1}(N)$ is a quotient of $(N/N')^{\otimes i}$. The induced derivation is the sum of the i commuting operators acting in individual tensor positions. If $D^s = 0$ on N/N', the sum has power $i(s-1)+1$ equal to zero, by the pigeonhole principle on its expanded products. The finite lower central filtration proves the assertion on N, in every characteristic.

Apply this fact to $N = W$. Since $\operatorname{ad} x$ maps L into W, it is nilpotent on L. Engel's theorem now gives nilpotency of L. The converse is immediate. If R is the solvable radical, L/R has zero center. Lemma 2.4 makes R invariant, and the result just proved applies to the restricted solution. □

4 The universal module and sharp metabelian bounds

4.1 A multigraded commutator module

For $d \geq 2$, put

$$R_d = \mathbb{F}_4[t_1, \dots, t_d]/(t_1^2, \dots, t_d^2), \qquad \mathfrak{m} = (t_1, \dots, t_d).$$

Let W_d be the R_d-module generated by v_{ij}, $i < j$, subject to

$$t_i v_{jk} + t_j v_{ik} + t_k v_{ij} = 0 \qquad (i < j < k). \tag{23}$$

We set $v_{ji} = v_{ij}$ and $v_{ii} = 0$. This is the classical free metabelian commutator-module presentation, base-changed to R_d; compare [9, Theorem 6.2] and [2, Sections 2–3].

Every derived module in Theorem 3.2 is a quotient of W_d: send t_i to P_i and v_{ij} to a_{ij}. The intrinsic-generation condition makes this homomorphism surjective. Conversely, every quotient of W_d, with these images, gives the normal form (10).

Grade R_d by coefficient degree, and put the generators v_{ij} in coefficient degree zero. We also use the finer grading $\deg t_i = e_i$, $\deg v_{ij} = e_i + e_j$.

Proposition 4.1. *The coefficient-degree Hilbert polynomial of* W_d *is*

$$H_d(T) = \sum_{k=2}^{d} \binom{d}{k} \left((k-1)T^{k-2} + kT^{k-1}\right) + \binom{d}{2} T^2. \tag{24}$$

Consequently

$$\dim_{\mathbb{F}_4} W_d = (d-1)2^d + 1 - d + \binom{d}{2}.$$

The universal solution on $\mathbb{F}_4^d \oplus W_d$ *has class* $\max(4, d+1)$.

Proof. A raw multihomogeneous monomial is $t_S v_{ij}$, where S is square-free. Its multidegree has entries zero, one, or two, and at most two entries are doubled. Fix its support of size $k \geq 2$.

If no index is doubled, the raw coordinates are the edges of the complete graph on the support. Relations (23) are its triangle sums. The quotient edge space has dimension $k-1$, in coefficient degree $k-2$.

If exactly one index j is doubled, only edges incident with j occur. Triangle relations through j identify every pair of these edges, leaving dimension one. There are k choices of j, in coefficient degree $k-1$.

If exactly two indices i, j are doubled, there is one raw coordinate. A third index in the support kills it by a triangle relation, since the other two terms contain a squared variable. The coordinate survives exactly when the support is $\{i, j\}$, giving the final term of (24). These cases exhaust the multigrading.

The maximum coefficient degree is two for $d = 2, 3$, and $d-1$ for $d \geq 4$. Equation (19) gives the class. □

In particular, binary degree two permits unbounded polar nilpotency class. For the exact minimum dimension we must retain a terminal commutator in as small a quotient of W_d as possible.

4.2 Terminal tensors and contraction modules

Lemma 4.2. *Let* $\mathfrak{l}$ *be a metabelian* $\mathbb{F}_4$*-Lie algebra satisfying* (15)*, of class* $c \geq 3$*. Some nonzero terminal commutator has a repeated input between its first bracket pair and an operator position. If* $c \geq 5$*, there is a subalgebra of the same class generated by exactly* $c-1$ *elements modulo its derived ideal.*

Proof. The last lower central ideal is spanned by

$$T(x_1, x_2; x_3, \ldots, x_c) = P(x_3) \cdots P(x_c)\beta(x_1, x_2).$$

Indeed $W = \operatorname{span} \beta + \mathfrak{m}W$, and multiplication by $\mathfrak{m}^{c-2}$ kills the second summand. The operator positions are symmetric, and a repeated input within them gives zero.

Suppose every value $T(x, y; x, z_4, \ldots, z_c)$ were zero. Polarizing the repeated input gives

$$T(x, y; z, \ldots) = T(z, y; x, \ldots).$$

Together with symmetry of the first pair and the operator positions, this makes T fully symmetric. Jacobi on the first three inputs then gives $3T = T = 0$, a contradiction.

Choose a nonzero repeated-input value. Its inputs span a subspace U_0 of the abelianization of dimension at most $c-1$. Their chosen lifts generate a subalgebra H of class c, with abelianization exactly U_0: all words of length at least two lie in its derived ideal. Its module is a quotient of W_{d_0}, where $d_0 = \dim U_0$. By Proposition 4.1, class $c \geq 5$ requires $d_0 \geq c-1$. Thus $d_0 = c-1$. □

For $d \geq 4$, the top coefficient degree of W_d is $d-1$. It has basis $T_1, \ldots, T_d$, where

$$T_j = \Big(\prod_{h \neq i} t_h\Big) v_{ij} \quad (i \neq j). \tag{25}$$

Triangle relations identify the choices of i.

Lemma 4.3. *The top component is the* $\mathrm{GL}(d,4)$*-module* $\det(U)\otimes U$*. For every nonzero functional* λ *on that component, extended by zero to the other degrees,*

$$\dim_{\mathbb{F}_4} R_d\lambda = 2^d - 2.$$

If a quotient M *of* W_d *retains a nonzero top component, then* $\dim_{\mathbb{F}_4} M \geq 2^d - 2$*, whether or not the quotient is graded.*

Proof. Under a diagonal basis change A, T_j scales by $\det(A)A_{jj}$. A transvection $e_j \mapsto e_j + \alpha e_k$ sends T_j to $T_j + \alpha T_k$, and fixes the other top basis vectors. To check the formula choose $i \notin \{j,k\}$ in (25); any extra product has a squared variable. For the remaining vectors an index outside the affected entries can be chosen since $d \geq 4$. Together with permutations these transformations prove the module assertion. They act transitively on nonzero dual vectors, including scalar multiples.

Normalize $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \neq j$. We claim

$$\mathrm{Ann}_{R_d}(\lambda) = \Big(\prod_{i\neq j} t_i\Big). \tag{26}$$

The annihilator is multigraded: contractions by distinct monomials have distinct resulting dual multidegrees. For a square-free monomial t_S not containing j, the remaining multidegree is of the one-doubled-index type. It survives unless S contains every other index. If $j \in S$, the remaining multidegree has no doubled index. It survives whenever an index is omitted, as witnessed by an edge joining that index to j; it vanishes when all indices are used. These are exactly the two monomials in the ideal on the right of (26). This proves the claim and the dimension formula.

For an arbitrary quotient M, pull back a functional nonzero on the top image to $\phi \in W_d^*$. Its highest homogeneous component ϕ_{d-1} is nonzero. Independent homogeneous contractions of ϕ_{d-1} lift to independent contractions of ϕ, by comparison of their leading dual degrees. Hence

$$\dim M \geq \dim R_d\phi \geq \dim R_d\phi_{d-1} = 2^d - 2.$$

This is the usual filtered contraction argument; it does not assume a graded kernel for $W_d \to M$. □

4.3 Exact minima

Theorem 4.4. *Among metabelian quadratic solutions of polar class* $c \geq 2$*, the minimum binary dimension is*

$$\mu(2) = 6, \quad \mu(3) = 8, \quad \mu(4) = 12, \qquad \mu(c) = 2^c + 2c - 6 \quad (c \geq 5).$$

Proof. Write $d = \dim_{\mathbb{F}_4}(V/W)$, $w = \dim_{\mathbb{F}_4} W$. Nonabelian objects have $d \geq 2$ by Lemma 2.6. A nonzero product $P_1 \cdots P_k v$, for commuting square-zero operators, yields 2^k independent vectors

$$v_S = \Big(\prod_{i\in S} P_i\Big)v \qquad (S \subseteq \{1,\dots,k\}).$$

In a relation, choose a term with nonzero coefficient of smallest support and multiply by its complementary operators. Every other surviving term repeats an operator and vanishes, while the chosen term does not. Taking $k = c - 2$ gives $w \geq 2^{c-2}$. For $c = 2,3,4$ this gives the respective lower bounds $6, 8, 12$.

They are attained by cyclic quotients of $W_2 = R_2 v_{12}$: use $R_2/\mathfrak{m}$, $R_2/(t_2)$, and R_2, respectively, as the derived module in (10). The first two representatives are

$$(a, b, z) \longmapsto (\omega a, \omega b, \omega z + ab),$$
$$(a, b, y, z) \longmapsto (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab).$$

The class-four representative appears in (36) below.

Let $c \geq 5$. Proposition 4.1 gives $d \geq c - 1$. Lemma 4.2 supplies a same-class subalgebra with $d_0 = c - 1$. Its derived module retains the top of W_{c-1}, so Lemma 4.3 gives $w \geq 2^{c-1} - 2$. Therefore

$$\dim_{\mathbb{F}_2} V = 2(d + w) \geq 2^c + 2c - 6.$$

To attain the bound, set $d = c - 1 \geq 4$, choose a top functional λ as in Lemma 4.3, and put

$$M_d = W_d / \{w : \lambda(rw) = 0 \text{ for every } r \in R_d\}. \tag{27}$$

The denominator is an R_d-submodule. The quotient has dimension $2^d - 2$ and retains the terminal value, so its solution has class $d + 1 = c$ and binary dimension $2(d + 2^d - 2)$. □

The change of formula at class five comes from the top degree of the universal module. For two and three quotient generators,

$$H_2(T) = 1 + 2T + T^2, \qquad H_3(T) = 3 + 8T + 6T^2.$$

In these cases the degree-two double-endpoint terms contribute at the top, allowing class four even with two quotient generators. For $d \geq 4$, they lie below the top degree $d - 1$; the terminal component $\det(U) \otimes U$ then controls class $d + 1$ uniformly. This is why the contraction argument yields one formula from class five onward, while the cyclic quotients of W_2 give the three smaller values.

The extremal quotient M_d in (27) has coefficient Hilbert polynomial

$$(d - 1) + \sum_{k=1}^{d-1} \binom{d}{d - 1 - k} T^k.$$

For example, the class-five extremizer has derived Hilbert vector $(3, 6, 4, 1)$ and binary lower central dimensions $(36, 28, 22, 10, 2, 0)$. The contraction construction supplies an explicit finite module description at every class, without an orbit search.

5 When the original polar bracket is $\mathbb{F}_4$-bilinear

Return to the normal form (10) and keep the intrinsic scalar actions on $U = V/W$ and W. Its auxiliary bracket (18) is always $\mathbb{F}_4$-bilinear. The following definition concerns the original polar bracket of f.

Definition 5.1. A metabelian solution is *compatible* if V admits an $\mathbb{F}_4$-vector space structure that restricts to the intrinsic structure on W, induces the intrinsic structure on U, and makes the original polar bracket $\mathbb{F}_4$-bilinear.

Theorem 5.2. *For a normal form with data P_i, a_{ij}, the following are equivalent:*

(i) *the solution is compatible;*

(ii) *there are $b_1, \ldots, b_d \in W$ satisfying*

$$P_i b_i = 0, \qquad P_i b_j = P_j a_{ij} \quad (i \neq j); \tag{28}$$

(iii) *the auxiliary $\mathbb{F}_4$-Lie algebra has a restricted square r with $r(W) = 0$ and $r(\mathfrak{l}) \subseteq W$.*

If a compatible scalar lift exists, the number of such scalar actions on the fixed binary space, with the prescribed actions on U, W, is 4^{dk}, where $k = \dim_{\mathbb{F}_4} \bigcap_i \ker P_i$.

Proof. Every binary-linear lift of multiplication by ω has the form

$$T(x, z) = (\omega x, \omega z + t(x)).$$

The relation $T^2 + T = \mathrm{id}$ is exactly $t(\omega x) = \omega^2 t(x)$. Thus $t(x) = b(x) = \sum_i x_i^2 b_i$. The involutory binary shear $S(x, z) = (x, z + b(x))$ conjugates T to standard scalar multiplication. It changes the cocycle of f to

$$Q + b + P(Q + b).$$

The first two terms have $\mathbb{F}_4$-bilinear polars. The last term has scalar weight zero. Its polar could be $\mathbb{F}_4$-bilinear only if that polar were zero, since an $\mathbb{F}_4$-bilinear polar has weight two. An additive zero-constant function of weight zero is zero: additivity and invariance under ω give its value at $\omega^2 x = x + \omega x$ equal to zero, and then invariance gives its value at x equal to zero. Compatibility is therefore equivalent to

$$P(x)(Q(x) + b(x)) = 0. \tag{29}$$

After the triangle cancellations, comparison of x_i^3 and $x_i x_j^2$ gives precisely (28).

We recall explicitly the restricted-square axioms in characteristic two: $r(\alpha u) = \alpha^2 r(u)$, $r(u+v) = r(u)+r(v)+[u, v]_0$, and $\operatorname{ad} r(u) = (\operatorname{ad} u)^2$. Any square with the range and vanishing conditions in (iii) must have the form

$$r(x, z) = (0, P(x)z + q(x)), \qquad q(x) = Q(x) + b(x). \tag{30}$$

Its first two axioms are automatic. For $u = (x, z), v = (y, t)$, the two adjoint expressions have final coordinates

$$P(x)P(y)z + P(x)\beta(x, y), \qquad P(x)P(y)z + P(y)q(x),$$

respectively. Condition (29) implies $P(x)\beta(x, y) = P(y)q(x)$: take the coefficient of s in $P(x + sy)q(x + sy) = 0$. Its degree in s is at most three, below $|\mathbb{F}_4|$, so the comparison is valid for functions over $\mathbb{F}_4$. This proves the third restricted axiom. Conversely that axiom, applied to u, gives $[u, r(u)]_0 = 0$, hence (29).

Finally two solutions of (28) differ by independent choices of d vectors in $\bigcap_i \ker P_i$. The displayed count follows. It counts scalar actions, not their conjugacy classes. □

Example 5.3. The eight-dimensional class-three model has $P_1 z = y$, $P_1 y = 0$, $P_2 = 0$, and $a_{12} = z$. Condition (28) would require $0 = P_2 b_1 = P_1 a_{12} = y$. Hence it has no compatible scalar action. This gives a concrete distinction between the auxiliary algebra and the original polar algebra.

Remark 5.4. The restricted square in Theorem 5.2 is a square on the auxiliary algebra. The theorem imposes both its range and its restriction to the derived ideal. General correspondences between quadratic maps and restricted Lie structures, such as [8, Section 9], carry their own additional identities. Here the exact additional condition is the solvable linear system (28).

6 Sharp compatible bounds and extremizers

6.1 The abelianization bound

Lemma 6.1. *A nonabelian compatible solution of class c, with $d = \dim_{\mathbb{F}_4}(V/W)$, satisfies $c \le d$.*

Proof. Use a restricted square from Theorem 5.2. The identity $P(x)\beta(x,y) = P(y)r(x)$, with a lift of x understood, implies

$$P(x)P(y)\beta(x,y) = 0. \tag{31}$$

In the universal module this kills the submodule K spanned by $t_i t_j v_{ij}$, $i < j$. It has dimension $\binom{d}{2}$ and lies in coefficient degree two. Every multiplication by an endpoint kills it by a square, and multiplication by a third index kills it by a triangle relation.

This relation space is invariant under $\mathrm{GL}(U)$. Indeed the universal expression $P(x)P(y)\beta(x,y)$ is additive in each input and scales quadratically in each. For example its additive cross term in x is

$$P(y)\big(P(x)\beta(z,y) + P(z)\beta(x,y)\big) = P(y)^2\beta(x,z) = 0$$

by Jacobi. Its basis values therefore span precisely K.

Put $\overline{W}_d = W_d/K$. The multigrading in Proposition 4.1 shows that its highest coefficient degree is $d-1$, for every $d \ge 2$, and that its top has the basis T_j of (25). This top is $\det(U)\otimes U$ also for $d = 2, 3$: the same diagonal and transvection calculations apply, with any additional double-endpoint term now in K. For $d = 2$ this is immediate from $T_1 = t_1 v_{12}, T_2 = t_2 v_{12}$.

Every compatible derived module M is a quotient of $\overline{W}_d$, so $c \le d+1$. Suppose $c = d+1$. Choose a functional on M nonzero on its terminal component. A change of quotient basis normalizes its top pullback to $\lambda(T_j) = 1$, $\lambda(T_k) = 0$ for $k \ne j$. Using $P_j a_{ij} = P_i b_j$, we have, for $i \ne j$,

$$\mathrm{im}(T_j) = \Big(\prod_{h\ne j} P_h\Big) b_j. \tag{32}$$

Write $b_j = \sum c_{kl} a_{kl} \pmod{\mathfrak{m}M}$, using intrinsic generation. The product on the right kills $\mathfrak{m}M$. If j is an endpoint of v_{kl}, then $(\prod_{h\ne j} t_h)v_{kl}$ is a top vector whose doubled endpoint is different from j. If j is not an endpoint, the vector has both endpoints doubled and is in K. Thus the normalized top functional is zero on the right side of (32), but one on the left. This contradiction proves $c \le d$. □

6.2 Counting preimages of an operator cube

Lemma 6.2. *A compatible solution of class $c \ge 2$ satisfies*

$$\dim_{\mathbb{F}_4} W \ge 2^{c-1} - 1.$$

Proof. The case $c = 2$ says $W \ne 0$. Let $c \ge 3$. By Lemma 4.2, choose a nonzero terminal value

$$P(x)P(z_1)\cdots P(z_{c-3})\beta(x,y).$$

Restrictedness gives $P(x)\beta(x,y) = P(y)b$, where $b = r(x) \in W$. Put $m = c-2$, $u_1 = y, u_2 = z_1, \ldots, u_m = z_{c-3}$. Then $P(u_1)\cdots P(u_m)b \ne 0$. The 2^m vectors

$$b_S = \Big(\prod_{i\in S} P(u_i)\Big) b \qquad (S \subseteq \{1,\ldots,m\})$$

are independent by the complementary-operator argument used in Theorem 4.4. Let C be their span. Since $[x, r(x)]_0 = 0$, $P(x)$ kills b, hence all of C.

For each nonempty S, choose $i \in S$ and set

$$v_S = \Big(\prod_{j \in S \setminus \{i\}} P(u_j) \Big) \beta(x, u_i).$$

These vectors lie in the intrinsic derived ideal, and restrictedness gives $P(x)v_S = b_S$. Thus

$$\dim \ker(P(x)|_W) \geq 2^m, \qquad \operatorname{rank}(P(x)|_W) \geq 2^m - 1.$$

Rank-nullity yields $\dim W \geq 2^{m+1} - 1 = 2^{c-1} - 1$. □

Theorem 6.3. *For every $c \geq 2$, the minimum binary dimension among compatible solutions of polar class c is*

$$\nu(c) = 2^c + 2c - 2.$$

Proof. Lemmas 6.1 and 6.2 give the lower bound by addition. For attainment put $s = c - 1$, $R_s = \mathbb{F}_4[t_1, \ldots, t_s]/(t_i^2)$, and use

$$f(x, z) = \left(\omega x, \omega z + \Big(\sum_i x_i t_i \Big) z \right) \quad \text{on } \mathbb{F}_4^s \oplus R_s. \tag{33}$$

The square of $\sum x_i t_i$ is zero, so direct substitution gives (1). The polar bracket is $\mathbb{F}_4$-bilinear and has intrinsic derived ideal $\mathfrak{m}R_s$. The scalar actions induced by f on that ideal and on the abelianization are the displayed ones, so the map is compatible. Its lower central ideals are $\gamma_k = \mathfrak{m}^{k-1} R_s$ for $k \geq 2$, giving class $s + 1 = c$. The derived and quotient dimensions are $2^s - 1$ and $s + 1$. The binary dimension is therefore $2(s + 2^s) = 2^c + 2c - 2$. □

For $c \geq 5$, the extra four binary dimensions have an exact structural meaning. The sharp bounds in the two subclasses are

	d	w
metabelian	$c - 1$	$2^{c-1} - 2$
compatible metabelian	c	$2^{c-1} - 1$

where each entry is the corresponding lower bound, attained simultaneously. Compatibility therefore requires one additional $\mathbb{F}_4$-dimension in the quotient and one in the derived ideal, giving four additional binary dimensions. The restricted square explains the strengthened bounds: it constrains terminal commutators and supplies independent preimages for the nonempty faces of the operator cube. The elementary semidirect construction attains this cost.

6.3 Several extremizers in every class

Theorem 6.4. *For every $c \geq 4$, at least $c - 2$ binary-linear conjugacy classes attain $\nu(c)$.*

Proof. Set $s = c - 1 \geq 3$ and

$$R = \mathbb{F}_4[t_0, \ldots, t_{s-1}]/(t_i^2), \qquad W = \mathfrak{m}R.$$

Let U have basis $x, u_1, \ldots, u_{s-1}, v$. Choose either $h = 0$, or $h = t_1 \cdots t_r$ with $1 \leq r \leq s - 2$, and put $p = t_0 h$. Define the action on W by multiplication:

$$P_x = t_0, \qquad P_{u_i} = t_i, \qquad P_v = p.$$

Define the alternating cocycle by

$$\beta(x, u_i) = t_i, \quad \beta(x, v) = t_0, \quad \beta(v, u_i) = (1 + h)t_i, \quad \beta(u_i, u_j) = 0.$$

The only Jacobi triple types to check are (x, u_i, u_j), (v, u_i, u_j), and (x, v, u_i). Their sums are, respectively,

$$t_i t_j + t_j t_i, \quad t_i(1+h)t_j + t_j(1+h)t_i, \quad t_0(1+h)t_i + pt_i + t_i t_0,$$

all zero. The cocycle values generate W under the action.

Take square values $b_x = t_0, b_{u_i} = 0, b_v = p$. They satisfy (28); the nontrivial relation involving p uses $p(1+h) = p$, since $h^2 = 0$. In scalar coordinates $a = (X, Y_1, \dots, Y_{s-1}, V)$, the resulting restricted square has the form

$$r(a, w) = (0, P(a)w + q(a)), \tag{34}$$

$$q(a) = t_0(X^2 + XV + hV^2) + \sum_i t_i Y_i (X + (1+h)V). \tag{35}$$

All ring elements in these formulas are fixed coefficients. Consequently $f = \omega \,\mathrm{id} + r$ has binary degree at most two. The square vanishes on W, takes values in W, and $[a, r(a)]_0 = 0$. Expanding $r(\omega a + r(a))$ proves $f^2 = f + \mathrm{id}$. Its original polar is the displayed $\mathbb{F}_4$-bracket, so it is compatible with the intrinsic scalar actions.

The lower central ideals are $\gamma_k = \mathfrak{m}^{k-1} R$ for $k \geq 2$, since the action includes every t_i. Each map has class c and dimension $\nu(c)$.

To distinguish them, use only intrinsic binary invariants. Let

$$\mathcal{A} = \mathrm{im}(L \longrightarrow \mathrm{End}_{\mathbb{F}_2}(W),\ a \longmapsto \mathrm{ad}\, a|_W), \qquad \mathcal{A}_k = \{D \in \mathcal{A} : D(W) \subseteq \gamma_{k+2}(L)\}.$$

For $h = 0$, $\dim_{\mathbb{F}_4} \mathcal{A} = s$ and $\mathcal{A}_k = 0$ for $k \geq 2$. For $h = t_1 \cdots t_r$, multiplication by p is independent of the linear multiplications and is nonzero on W, as $r \leq s-2$. Thus $\dim_{\mathbb{F}_4} \mathcal{A} = s+1$, and

$$\dim_{\mathbb{F}_4} \mathcal{A}_k = \begin{cases} 1, & 2 \leq k \leq r+1, \\ 0, & k \geq r+2. \end{cases}$$

Indeed any nonzero linear multiplication has a nonzero degree-two output on W, whereas pW has minimum nonzero degree $r+2$. The binary dimensions are twice these dimensions and are preserved by every binary-linear conjugacy, independently of a choice of scalar lift. They separate $h = 0$ and the $s-2$ possible values of r. □

The theorem gives an explicit collection of extremizers. It leaves open the classification of all equality cases and their automorphism groups.

7 Two closed classification ranges

The polar framework supports two complementary classification questions: fix the ambient dimension, or fix the abelianization while allowing the ambient dimension to grow. The first application combines simple-algebra obstructions with the metabelian correspondence to close dimension eight. The second uses central descent and exact extension spaces to close the entire solvable abelianization-four stratum. Its last two members are nonmetabelian, so the extension method also reaches beyond the correspondence of Section 3.

7.1 Obstructions to simple polar algebras

The simple reduction in Corollary 2.5 can be combined with two explicit obstructions.

Proposition 7.1. *No polar algebra has a Lie subquotient isomorphic to the underlying binary Lie algebra of $\mathfrak{sl}_3(K)$, for a finite field K of characteristic two. In particular it cannot be $\mathfrak{sl}_m(K)$ or $\mathfrak{psl}_m(K)$, $m \geq 3$.*

Proof. In $\mathfrak{sl}_3(K)$, take $a = E_{12}, b = E_{23}, c = E_{31}, d = E_{11} + E_{22}$. The three terms in (7) are $E_{11} + E_{33}, E_{22} + E_{33}, 0$, whose sum is nonzero. A polynomial identity passes to subalgebras and quotients. The upper 3×3 block gives the required subalgebra in $\mathfrak{sl}_m(K)$. For $m \geq 4$ it also embeds in the quotient by scalar matrices, since the block intersects that center trivially. For $m = 3$ the center is zero. □

Lemma 7.2 (Two centralizers)**.** *Let L be a Lie algebra over a characteristic-two field K. If $x, y \in L$ satisfy*

$$C_L(x) = Kx, \qquad C_L(x + y) = K(x + y), \qquad [y, [x, y]] \notin K[x, y],$$

then there is no zero-constant binary-quadratic map with polar bracket $[\ ,\ ]$ satisfying $[v, f(v)] = 0$ for all v.

Proof. The commuting condition forces $f(x) = \alpha x$, $f(x + y) = \beta(x + y)$, for some $\alpha, \beta \in K$. Polarization then gives $f(y) = (\alpha+\beta)x+\beta y+[x, y]$. Commuting this value with y contradicts the third hypothesis. No K-linearity of f is used. □

Proposition 7.3. *For every finite characteristic-two field K and $r \geq 2$, the derived divided-power Witt algebra $W(1; r)'$ over K, viewed over $\mathbb{F}_2$, is not a polar algebra of a quadratic solution.*

Proof. Use the classical divided-power algebra $A = O(1; r)$, with basis $x^{(a)}$, $0 \leq a < 2^r$, and multiplication

$$x^{(a)}x^{(b)} = \binom{a+b}{a} x^{(a+b)}$$

when $a + b < 2^r$, and zero otherwise. Let $D(x^{(a)}) = x^{(a-1)}$, with $D(1) = 0$. Then $\ker D = K$, and $W(1; r)' = \operatorname{span}_K\{x^{(a)}D : 0 \leq a \leq 2^r - 2\}$; see [6, Section 4.5].

If u is a unit of A, a commuting vector vD satisfies $u^2D(v/u) = 0$, so $C_{AD}(uD) = KuD$. Choose $x = D, y = x^{(2)}D$. Both 1 and $1 + x^{(2)}$ are units, so the first two hypotheses of Lemma 7.2 hold also inside $W(1; r)'$. Finally

$$[x, y] = x^{(1)}D, \qquad [y, [x, y]] = x^{(2)}D,$$

because $(x^{(1)})^2 = 0$. These are K-independent. □

7.2 All eight-dimensional solutions

Theorem 7.4. *In binary dimension eight there are exactly three linear-conjugacy classes of quadratic solutions. Representatives on $\mathbb{F}_4^4$ are*

$$\begin{aligned} L(a, b, c, d) &= (\omega a, \omega b, \omega c, \omega d), \\ C(a, b, c, d) &= (\omega a, \omega b, \omega c, \omega d + ab), \\ N(a, b, y, z) &= (\omega a, \omega b, \omega y + az + a^2 b, \omega z + ab). \end{aligned}$$

Their polar-image dimensions are $0, 2, 4$, their center dimensions are $8, 4, 2$, and their classes are $1, 2, 3$, respectively.

Proof. We first exclude nonsolvable polar algebras in every binary dimension at most eight. Corollary 2.5 would give an even-dimensional nonabelian simple polar algebra of dimension at most eight. The classical lists of Strade [11] and Vaughan-Lee [12, Theorem 1] leave the six-dimensional algebra obtained from $W(1; 2)'$ over $\mathbb{F}_4$, and the two eight-dimensional simple types over $\mathbb{F}_2$. The first is excluded by Proposition 7.3.

For the other two, use the published bases $b_1, \ldots, b_8$ in the two tables of [3, Section 3.6]. The three terms of (7), at (b_1, b_2, b_3, b_4), are as follows:

Type	$[[b_1, b_2], [b_3, b_4]]$	$[[b_1, b_3], [b_2, b_4]]$	$[[b_1, b_4], [b_2, b_3]]$
First	0	0	b_8
Second	$b_2 + b_3 + b_4$	$b_2 + b_3$	$b_2 + b_4$

Both sums are nonzero. The list is an external classification input; these short evaluations supply the equation-specific exclusions.

Thus all polar algebras in the stated dimensions are nilpotent by Theorem 3.4. A nonabelian nilpotent polar algebra has dimension at least six, by Lemma 2.6 and parity of its nonzero derived ideal. In dimension eight its derived ideal has dimension at most four and is therefore abelian. The metabelian correspondence applies.

If $W = 0$, there is the single linear class. If $\dim_{\mathbb{F}_4} W = 1$, all square-zero actions vanish; the cocycle is an alternating scalar form on $\mathbb{F}_4^3$. Its nonzero rank is two, giving C. If $\dim_{\mathbb{F}_4} W = 2$, the quotient has dimension two. A zero action could not generate W from its single alternating coefficient. For a nonzero square-zero action P_1, its commuting square-zero partners are scalar multiples of it. Change quotient basis so $P_2 = 0$, and normalize $a_{12} = z, P_1 z = y, P_1 y = 0$. This gives N. The displayed intrinsic dimensions distinguish the three classes. □

7.3 Central descent with the smallest nontrivial abelianization

We now assume solvability and fix $\dim_{\mathbb{F}_2}(V/W) = 4$. This fixes the smallest abelianization dimension of a nonabelian nilpotent polar algebra. The following theorem closes the stratum in all ambient dimensions.

Theorem 7.5 (A closed seven-class chain)**.** *Among solvable polar solutions with binary abelianization dimension four, there is exactly one conjugacy class in each of dimensions* $4, 6, 8, 10, 12, 14, 16$*, and no class in any other dimension. Their polar classes are* $1, 2, 3, 3, 4, 5, 5$*. The extension-space ranks in the proof are exact finite computations, certified as described in Appendix A.*

The representatives are the successive prefixes of the following map, with coordinates ordered as (a, b, c, d, e, h, r, s):

$$\begin{aligned}
a' &= \omega a, & b' &= \omega b,\\
c' &= \omega c + ab,\\
d' &= \omega d + ac + a^2 b,\\
e' &= \omega e + bc + ab^2,\\
h' &= \omega h + ae + bd,\\
r' &= \omega r + A, & A &= ac^2 + a^2 e + ah + cd,\\
s' &= \omega s + B, & B &= bc^2 + b^2 d + bh + ce.
\end{aligned} \tag{36}$$

Each prefix is binary-quadratic. Every new cocycle q satisfies $q(g(x)) = \omega^2 q(x)$ for the preceding prefix g, proving (1) recursively. For the last two expressions this follows by substitution; the extra terms cancel in pairs or have coefficient $1 + \omega + \omega^2 = 0$.

Proof of Theorem 7.5. The proof proceeds from descent to extensions and then to closure. Every member descends to the abelian four-dimensional solution. At each prefix we compute the complete quadratic extension quotient; kernel scalings and quotient automorphisms put all its nonzero vectors into a single orbit. This gives a unique next class until the final extension quotient vanishes. Descent then turns that last calculation into a bound on every ambient dimension.

Table 2: Intrinsic invariants of the seven prefixes. All dimensions are binary.

n	$\dim W$	$\dim Z$	Class	Lower central dimensions
4	0	4	1	$(4, 0)$
6	2	2	2	$(6, 2, 0)$
8	4	2	3	$(8, 4, 2, 0)$
10	6	4	3	$(10, 6, 4, 0)$
12	8	2	4	$(12, 8, 6, 2, 0)$
14	10	2	5	$(14, 10, 8, 4, 2, 0)$
16	12	4	5	$(16, 12, 10, 6, 4, 0)$

Descent. Solvability implies nilpotency. In a nonabelian member of this stratum, $Z(L) \subseteq W$. Otherwise the linear scalar action on $Z(L)$ would give a central invariant $\mathbb{F}_4$-line disjoint from W. Its quotient would be a nonabelian nilpotent polar algebra with binary abelianization dimension two, contradicting Lemma 2.6. Quotient by any central $\mathbb{F}_4$-line in W. The dimension drops by two, the abelianization dimension stays four, and a quadratic solution is inherited. Repetition ends at the unique abelian four-dimensional solution. Every member is therefore reached by successive central $\mathbb{F}_4$-extensions.

The complete extension space. For a fixed quotient solution g, every such extension has the form

$$F(x, z) = (g(x), \omega z + q(x)), \qquad q(g(x)) = \omega^2 q(x), \quad q(0) = 0,$$

where q is binary-quadratic. Let E_g be this $\mathbb{F}_4$-vector space and define

$$H_g = E_g / \delta \operatorname{Hom}_{\mathbb{F}_2}(V, \mathbb{F}_4), \qquad \delta(t) = t \circ g + \omega t. \tag{37}$$

This quotient accounts for every binary-linear change of splitting. Kernel scalings and quotient automorphisms act by scaling and pullback. The zero class splits off a central scalar line and increases the binary abelianization dimension to six. Conversely, if an extension has its new line disjoint from its derived ideal, choose an $\mathbb{F}_4$-complement to that line in the abelianization and take its inverse image. It is an invariant complement containing the derived ideal, so the extension splits and its class in H_g is zero. Thus exactly the nonzero classes remain in our stratum.

Ranks and representatives. The complete results are in Table 3. The equation for q has binary degree at most four, so values at inputs of binary weight at most four recover every coefficient by Boolean Möbius inversion. The certificates verify the ranks and the spanning of each displayed quotient. The splitting-image dimension also follows directly: $\delta(t) = 0$ precisely for the linear maps t that annihilate W and intertwine the intrinsic ω-actions. This kernel has binary dimension four, so $\dim \operatorname{im} \delta = 2n - 4$.

Table 3: Exact central-extension spaces. Here n is the dimension of g.

n	Unknowns	Rank	$\dim_{\mathbb{F}_2} E_g$	$\dim_{\mathbb{F}_2} \operatorname{im} \delta$	$\dim_{\mathbb{F}_4} H_g$	Basis of H_g
4	20	14	6	4	1	ab
6	42	30	12	8	2	$ac + a^2b,\ bc + ab^2$
8	72	58	14	12	1	$bc + ab^2$
10	110	92	18	16	1	$ae + bd$
12	156	132	24	20	2	A, B
14	210	184	26	24	1	B
16	272	244	28	28	0	–

Projective orbits. When H_g is one-dimensional, kernel scalings identify all its nonzero classes. There are two two-dimensional cases. For $n = 6$, put $u = ac + a^2b, v = bc + ab^2$. The following automorphisms of g act on their classes by a swap, a nontrivial diagonal scaling, and a shear:

$$\begin{aligned}
&(a, b, c) \mapsto (b, a, c),\\
&(a, b, c) \mapsto (\lambda a, b, \lambda c), \qquad \lambda \in \mathbb{F}_4^*,\\
&(a, b, c) \mapsto (a + b, b, c + b^2).
\end{aligned}$$

Their actions are $(u, v) \mapsto (v, u)$, $(\lambda^2 u, \lambda v)$, and $(u + v, v)$. They are transitive on the five projective points of $\mathbb{F}_4^2$.

For $n = 12$, the corresponding automorphisms are

$$\begin{aligned}
&(a, b, c, d, e, h) \mapsto (b, a, c, e, d, h),\\
&(a, b, c, d, e, h) \mapsto (\lambda a, b, \lambda c, \lambda^2 d, \lambda e, \lambda^2 h),\\
&(a, b, c, d, e, h) \mapsto (a + b, b, c + b^2, d + e, e, h).
\end{aligned}$$

They send (A, B) to (B, A), $(A, \lambda^2 B)$, and $(A + B + ab + b^2, B + b^2)$. The extra terms are splitting changes: $ab = \delta(c)$, $b^2 = \delta(b^2)$. The projective action is again transitive.

Closure. Starting with the abelian four-dimensional map, every nonzero extension class therefore yields the unique next prefix, through dimension sixteen. At that point $H_g = 0$, so no next member exists. Descent rules out every higher dimension as well. The coordinate brackets give Table 2. Their second derived dimensions in dimensions fourteen and sixteen are two and four; these are the two nonmetabelian members. □

The six-dimensional prefix is the central example already present in [14, Lemma 10.3(1)]. General central recursion also appears there. The quadratic eigenspaces, linear-conjugacy orbits, and the vanishing of the final extension quotient are the ingredients that close the present seven-class range.

8 Further structural questions

The structure theory suggests three directions, each with a concrete starting point and a different obstruction to overcome.

Can a nonsolvable polar algebra occur? Corollary 2.5 reduces this to a simple polar example. Such an algebra must have even binary dimension, satisfy (7), and admit a commuting quadratic refinement. The matrix and Witt obstructions give two tests for candidate algebras. A general obstruction would settle the question negatively; a simple polar example would produce solutions beyond the nilpotent regime. The main issue is to construct or exclude a quadratic refinement satisfying the full functional equation.

What is the dimension cost beyond the metabelian case? The fourteen-dimensional member of Theorem 7.5 already has class five, whereas the metabelian minimum at that class is thirty-six. Thus the all-solution minimum at class five is at most fourteen, and lower bounds for that problem must account for nonabelian derived ideals. The closed abelianization-four stratum supplies exact reference cases. A general theory would need to control further extension patterns as the abelianization grows, beyond the commuting module actions used here.

Which structures attain the sharp metabelian bounds? Theorem 6.4 gives at least $c-2$ compatible equality cases in class $c \geq 4$. Classifying all equality cases requires determining which module quotients and cocycles attain the bounds, and when their resulting maps are conjugate. The contraction quotients in (27), the filtered action spaces $\mathcal{A}_k$, and the morphism correspondence provide tools for this problem and for computing the extremizers' automorphism groups.

There is also a geometric interpretation of every result: nonzero orbits define an oriented binary line spread, and linear conjugacy preserves the underlying additive geometry. The polar Lie algebra records degree-two interactions between those oriented lines. Its center has a direct functional meaning. For $a \in Z(L)$,

$$f(x+a)+f(x)=f(a) \qquad \text{for every } x.$$

Consequently nilpotent models have constant differences in nonzero central directions. Their structure is particularly suited to the algebraic and geometric classification questions developed here.

A Exact certificates for the central-extension ranks

This appendix specifies the finite rank proof in Theorem 7.5. The general correspondence and both all-class extremal bounds are proved independently in the main text.

A.1 Encoding and equations

Use the binary basis $1, \omega$ of $\mathbb{F}_4$, with $\omega^2 = \omega + 1$. For an n-dimensional binary domain, the nonconstant ANF monomials of degree at most two are x_i and $x_i x_j$, $i < j$. A candidate $\mathbb{F}_4$-valued q therefore has $2(n + \binom{n}{2}) = n(n+1)$ binary unknowns. The map g is stored by the binary output vector of each such monomial, with the scalar-coordinate order of (36).

For each binary point x, evaluating $q(g(x)) + \omega^2 q(x)$ gives two binary rows in these unknowns. Their degree is at most four. Requiring all rows at inputs of weights one through four is equivalent to requiring the entire function to vanish: the value at zero is already zero, and triangular Boolean Möbius inversion recovers every coefficient of degree at most four.

A.2 What each certificate proves

For each $n \in \{4, 6, 8, 10, 12, 14, 16\}$, the supplied certificate contains:

(i) the quadratic coefficients of the quotient map and the monomial order;

(ii) a stated number of independent rows, each with its originating input point and output bit;

(iii) a kernel basis annihilated by every weight-at-most-four row;

(iv) the actual vectors $\delta(t)$ for a basis of binary-linear changes of splitting;

(v) the named representatives of H_g in Table 3.

The independent checker reconstructs the rows from the stored map, checks row independence over $\mathbb{F}_2$, checks the entire kernel on all required inputs, and verifies that row rank plus kernel dimension is the number of unknowns. It then reconstructs the splitting-change vectors and verifies that their span together with the stated quotient representatives is exactly the kernel. Thus both the upper and lower rank bounds are certified.

These are exact, reproducible algebraic checks; they are not proof-assistant formalizations.

A.3 Files and replay

The `verification` directory contains the seven certificates, their SHA-256 manifest, and three checkers. From the source-project directory, run:

```
cd verification
python check_chain_rank_certificates.py
python audit_metabelian_symbolic_interfaces.py
python manuscript_transcription_checks.py
```

Use Python 3 and, for the first program, SymPy. The rank checker authenticates each record, rebuilds the equations from the stored map, and uses exact $\mathbb{F}_2$ linear algebra in SymPy's domain-matrix implementation; it imports no generator code. The other programs use the standard library: the second checks scalar weights, binary degree, and morphism interfaces; the third checks coordinate transcription, intrinsic dimensions, and projective actions. Each prints JSON results with status `PASS` and writes reports under `verification`. Expected ranks and quotient dimensions are in Table 3.

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